



**SCHOOL OF COMPUTER SCIENCE ENGINEERING AND INFORMATION
SYSTEMS**

Winter Semester 2025-26

Agile Development Process and DevOps

ISWE406L

C2+TC2

ASSIGNMENT – 4: DOCKER

Faculty:

Prof. Dr. Sweta Bhattacharya

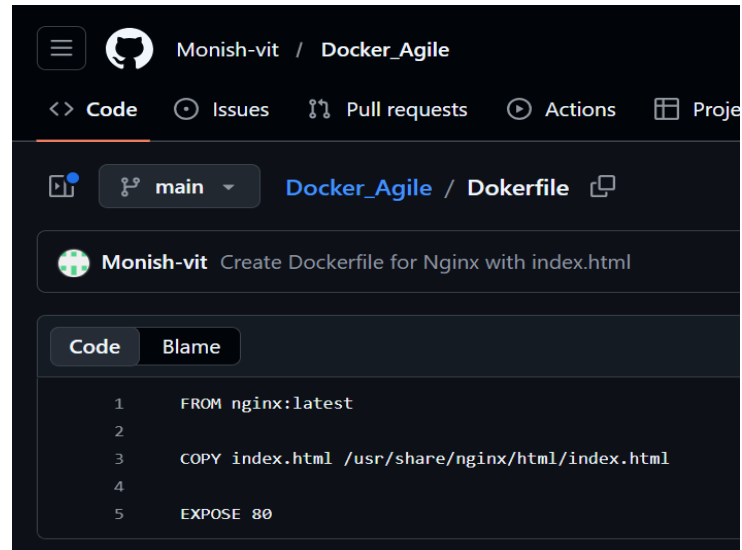
Submitted By:

S Monishwar

23MIS0690

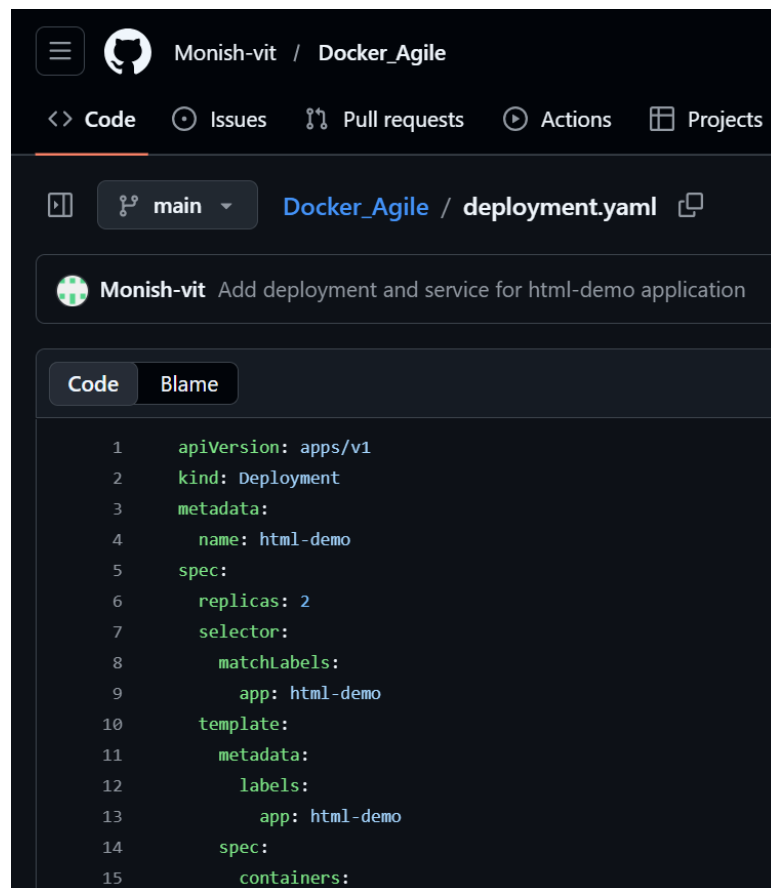
1. Create a repository in GIT and create the following files:

- index.html
- Dockerfile
- Deployment.yaml



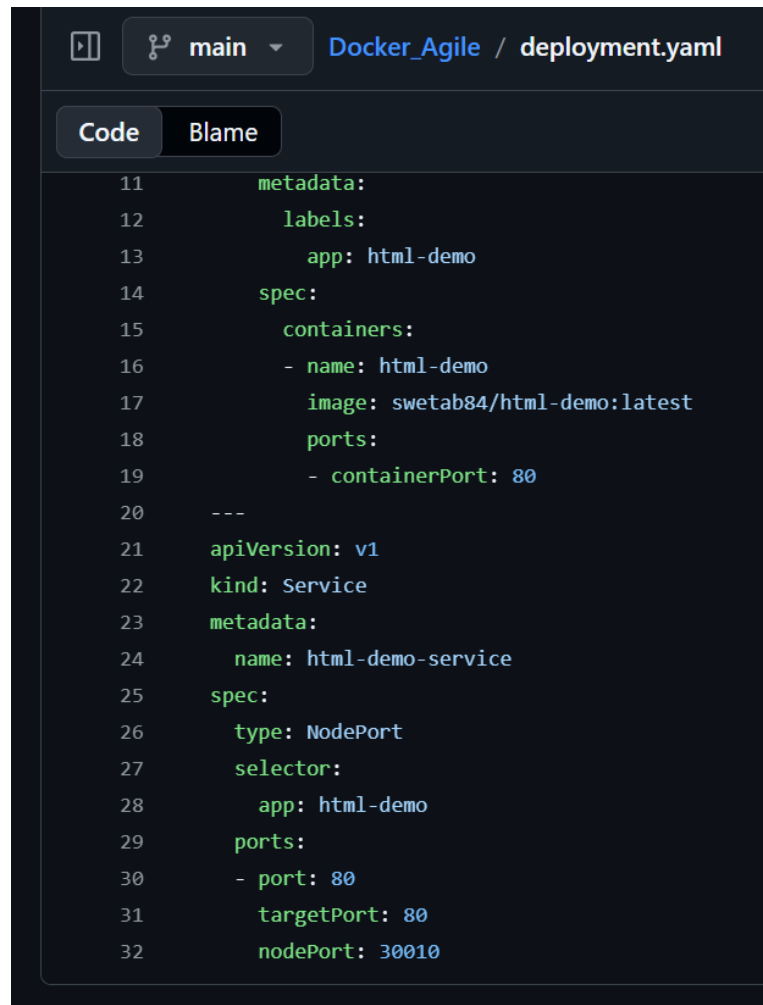
The screenshot shows the GitHub interface for the repository 'Monish-vit / Docker_Agile'. The 'main' branch is selected, and the file 'Dockerfile' is open. The commit message is 'Monish-vit Create Dockerfile for Nginx with index.html'. The code content is as follows:

```
1 FROM nginx:latest
2
3 COPY index.html /usr/share/nginx/html/index.html
4
5 EXPOSE 80
```



The screenshot shows the GitHub interface for the repository 'Monish-vit / Docker_Agile'. The 'main' branch is selected, and the file 'deployment.yaml' is open. The commit message is 'Monish-vit Add deployment and service for html-demo application'. The code content is as follows:

```
1 apiVersion: apps/v1
2 kind: Deployment
3 metadata:
4   name: html-demo
5 spec:
6   replicas: 2
7   selector:
8     matchLabels:
9       app: html-demo
10  template:
11    metadata:
12      labels:
13        app: html-demo
14    spec:
15      containers:
```



```
11     metadata:
12     labels:
13         app: html-demo
14     spec:
15     containers:
16     - name: html-demo
17       image: swetab84/html-demo:latest
18     ports:
19     - containerPort: 80
20 ---
21 apiVersion: v1
22 kind: Service
23 metadata:
24     name: html-demo-service
25 spec:
26     type: NodePort
27     selector:
28         app: html-demo
29     ports:
30     - port: 80
31       targetPort: 80
32       nodePort: 30010
```

Index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Monishwar S Portfolio</title>
<style>
body {
margin: 0;
font-family: 'Segoe UI', sans-serif;
background-color: #f5f7fa;
color: #333;
}
header {
background: #2c3e50;
color: white;
text-align: center;
```

```
padding: 30px 10px;
}
header h1 {
margin: 5px;
}
header p {
margin: 5px;
font-size: 18px;
}
nav {
background: #34495e;
padding: 10px;
text-align: center;
}
nav a {
color: white;
margin: 0 15px;
text-decoration: none;
font-weight: bold;
}
nav a:hover {
text-decoration: underline;
}
section {
padding: 40px 20px;
max-width: 900px;
margin: auto;
}
h2 {
color: #2c3e50;
border-bottom: 2px solid #ddd;
padding-bottom: 5px;
}
.card {
background: white;
padding: 20px;
margin-top: 20px;
border-radius: 8px;
box-shadow: 0 2px 5px rgba(0,0,0,0.1);
}
ul {
padding-left: 20px;
}
footer {
text-align: center;
background: #2c3e50;
color: white;
padding: 10px;
```

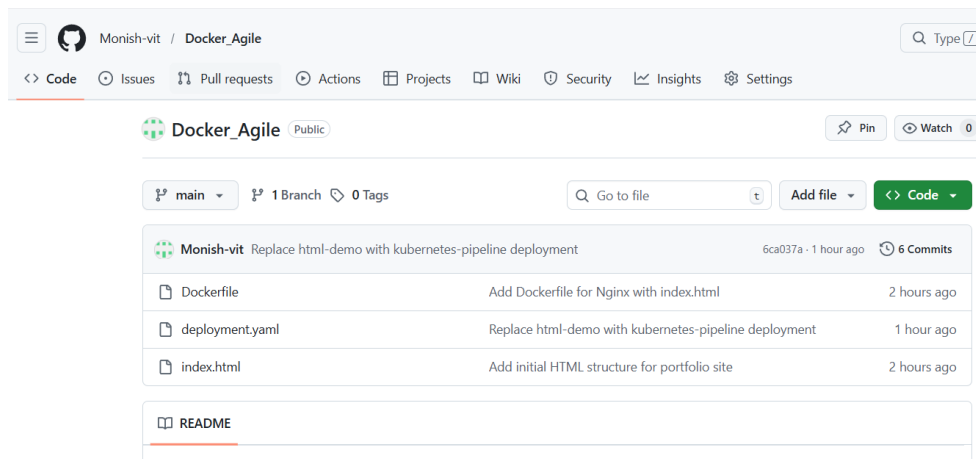
```

}
button {
padding: 10px 15px;
border: none;
background: #3498db;
color: white;
cursor: pointer;
border-radius: 5px;
}
button:hover {
background: #2980b9;
}
</style>
</head>
<body>
<header>
<h1>Monishwar S</h1>
<h3>23MIS0690</h3>
<p>Software Engineering Student | VIT University, Vellore</p>
</header>
<nav>
<a href="#summary">Summary</a>
<a href="#experience">Experience</a>
<a href="#projects">Projects</a>
<a href="#contact">Contact</a>
</nav>
<section id="summary">
<h2>Professional Summary</h2>
<div class="card">
<p>
Enthusiastic Software Engineering student at VIT University with a strong interest in
full-stack development, data analytics, and machine learning. Passionate about building
efficient, scalable solutions and solving real-world problems through technology.
Skilled in Python, Java, and web technologies with hands-on experience in academic
and personal projects.
</p>
</div>
</section>
<section id="experience">
<h2>Experience</h2>
<div class="card">
<ul>
<li><b>Web Development Intern</b> – TechSolutions Pvt Ltd (2025)<br>
Developed responsive web pages using HTML, CSS, and JavaScript.</li>
<li><b>Project Team Member</b> – Academic Project (2024–2025)<br>
Collaborated in a team of 4 to develop a web-based system using Python and MySQL.</li>
<li><b>Hackathon Participant</b> – VIT Hack 2025<br>
Built a prototype AI-based solution for real-time data analysis.</li>

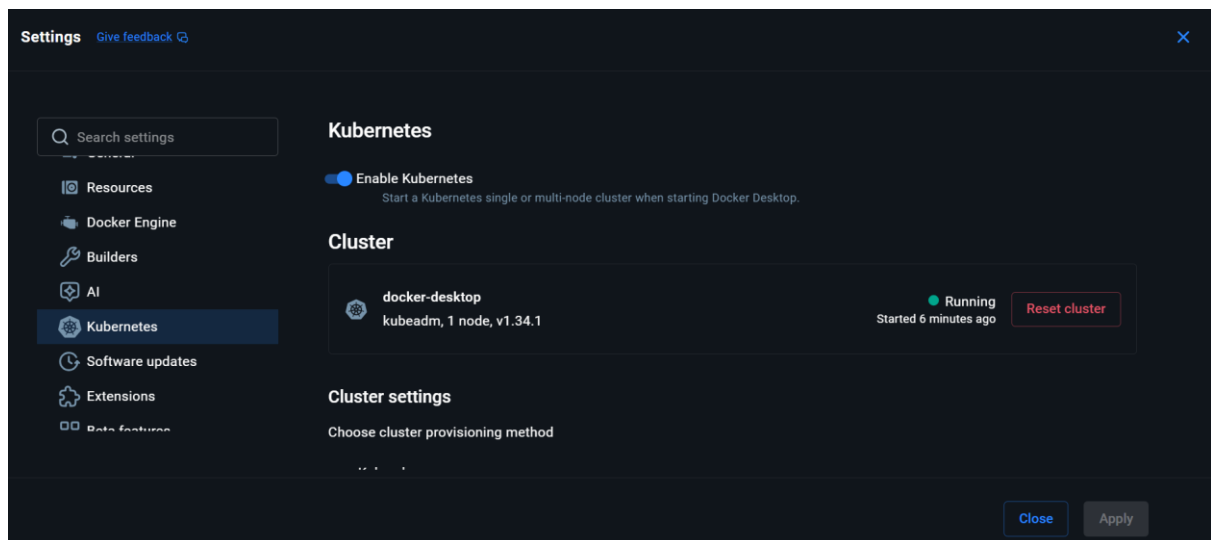
```

```
</ul>
</div>
</section>
<section id="projects">
<h2>Projects</h2>
<div class="card">
<ul>
<li><b>AI-Based Crime Prediction System</b><br>
Developed a machine learning model to predict crime patterns using historical data.</li>
<li><b>Railway Reservation System</b><br>
Designed a web-based system with deadlock avoidance algorithm.</li>
<li><b>Student Performance Predictor</b><br>
Built a predictive model to analyze student outcomes based on academic data.</li>
</ul>
<button onclick="showProjects()">Show Project Message</button>
</div>
</section>
<section id="contact">
<h2>Contact Details</h2>
<div class="card">
<p>Email: monishwar@example.com</p>
<p>Phone: +91 98765 43210</p>
<p>LinkedIn: linkedin.com/in/monishwar</p>
<button onclick="contactMsg()">Contact Me</button>
</div>
</section>
<footer>
<p>&copy; 2026 Monishwar S Portfolio</p>
</footer>
<script>
function showProjects() {
alert("Projects showcase innovation and practical implementation!");
}
function contactMsg() {
alert("Thank you for reaching out! I will get back to you soon.");
}
</script>
</body>
</html>
```

Github Repository



2. Enable Kubernetes



<

Add Username with password

×

Scope ?
Global (Jenkins, nodes, items, all child items, etc) ▼

Username
monishwar

☐ Treat username as secret ?

Password
.....

ID ?
dockerhub

Create

< Add Secret file

×

Scope ?
Global (Jenkins, nodes, items, all child items, etc) ▼

File

Choose File

 config

ID ?
kubernetes

Description ?

Create

Create Pipeline in Jenkins:

Jenkins

kubernetes-pipeline

Configure

Q ⚙️ 👤

Configure

General

Triggers

Pipeline

Advanced

Pipeline script ▼

Script ?

try sample Pipeline...

```
1 pipeline {
2   agent any
3   environment {
4     DOCKER_IMAGE = "monishwar-kubernetes-pipeline"
5   }
6   stages {
7     stage('Clone Code') {
8       steps {
9         git branch: 'main',
10        url: 'https://github.com/Monish-vit/Docker_Agile.git'
11      }
12    }
13    stage('Build Docker Image') {
14      steps {
15        bat 'docker build -t %DOCKER_IMAGE% .'
```

Save Apply

Jenkins Output:

```
service/kubernetes-pipeline-service created

[Pipeline] bat

C:\ProgramData\Jenkins\jenkins\workspace\kubernetes-pipeline>kubect1 get pods

NAME                                READY   STATUS    RESTARTS   AGE
kubernetes-pipeline-5f9664dfcc-9816t  0/1     ContainerCreating  0          1s
kubernetes-pipeline-5f9664dfcc-vn5bb  0/1     ContainerCreating  0          1s

[Pipeline] bat

C:\ProgramData\Jenkins\jenkins\workspace\kubernetes-pipeline>kubect1 get svc

NAME                                TYPE        CLUSTER-IP    EXTERNAL-IP  PORT(S)    AGE
kubernetes                         ClusterIP   10.96.0.1     <none>       443/TCP    3d21h
kubernetes-pipeline-service        NodePort    10.109.69.49  <none>       80:30010/TCP 1s

[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

Jenkins / kubernetes-pipeline

Status

Changes

Build Now

Configure

Delete Pipeline

Stages

Rename

Pipeline Syntax

kubernetes-pipeline

Add description

Permalinks

- Last build (#3), 1 hr 45 min ago
- Last stable build (#3), 1 hr 45 min ago
- Last successful build (#3), 1 hr 45 min ago
- Last failed build (#2), 1 hr 49 min ago
- Last unsuccessful build (#2), 1 hr 49 min ago
- Last completed build (#3), 1 hr 45 min ago

In Docker:

Containers

- Containers
- Images
- Volumes
- Kubernetes
- Builds
- Models
- MCP Toolkit BETA
- Docker Hub
- Docker Scout
- Extensions

Containers [Give feedback](#)

Container CPU usage ⓘ

0.00% / 1200% (12 CPUs available)

Container memory usage ⓘ

27.65MB / 7.42GB

[Show charts](#)

☰ ● Only show running containers

☐		Name	Container ID	Image	Port(s)	CPU (%)	Mem	Actions
☐	●	k8s_kubernetes	a617f29f27ea	kamaleshw		0%	17.2	⚙️ 🟦 ⋮ 🗑️
☐	●	k8s_kubernetes	98b6a5b724d6	kamaleshw		0%	10.3	⚙️ 🟦 ⋮ 🗑️

In LocalHost: HTML script output.

